

FPGA Tick-to-Trade Engine

A nanosecond-latency, fully-pipelined hardware trading datapath for NASDAQ market data

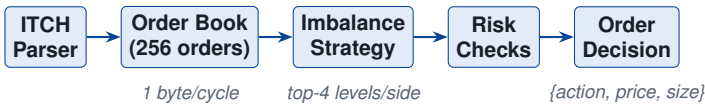
SystemVerilog Xilinx UltraScale+ Vivado (synth + P&R) cocotb / Python AXI4-Stream / AXI4-Lite NASDAQ ITCH 5.0 fixed-point DSP

205 ns	0 ns	200 MHz	200 MB/s
tick-to-trade latency	jitter (deterministic)	timing-closed (WNS > 0)	line-rate ITCH ingest

Overview

An end-to-end high-frequency-trading pipeline implemented entirely in FPGA fabric: it ingests a raw NASDAQ **ITCH 5.0** byte stream, maintains a live **limit order book**, computes a depth-weighted **imbalance trading signal**, applies **pre-trade risk checks**, and emits an order decision in a fixed **205 ns** (41 clock cycles at 200 MHz). The datapath is a deeply pipelined spatial dataflow with **zero floating-point** and a hardware latency counter for self-measurement. A bit-accurate Python model mirrors the hardware exactly, serving both as the verification golden reference and as the software performance baseline.

Architecture — 5-stage hardware pipeline



Streams 1 ITCH byte per clock; a 256-slot direct-mapped order table with a sorted top-4-levels-per-side cache; depth-weighted imbalance signal with spread guard and imbalance-scaled lot sizing; priority-encoded risk engine (order-size, price-band, net position, kill-switch). Host I/O over AXI4-Stream (data) and AXI4-Lite (control + on-chip latency histogram readout).

Results & performance vs. software

Metric	Detail	Value
Tick-to-trade latency	41 cycles @ 200 MHz, parse-to-decision	205 ns
Determinism	fixed-latency pipeline, no caches/OS/scheduler	0 ns jitter
Speed-up vs optimized C	same computation, median (compiled -O2)	~5×
Tail-latency advantage	CPU spikes to ~1 ms; FPGA fixed at 205 ns	up to ~1,800×
Throughput	1 byte/cycle ITCH ingest	200 MB/s
Timing closure	Vivado P&R, UltraScale+, positive worst slack	200 MHz
Logic footprint	single-symbol engine	~7K LUTs
ITCH message types	A, F, C, D, E, U, X, P	8 decoded

Verification & engineering rigor

- **81/81** cocotb tests passing; every RTL block diffed against a bit-accurate Python golden model on large synthetic and replayed ITCH streams.
- **Latency self-measured** on-chip (cycle-accurate counter, read over AXI4-Lite) and corroborated by Vivado static timing analysis.
- Software baselines (compiled C and Python) implemented for an apples-to-apples latency comparison of the identical computation; SystemVerilog assertions and back-pressure no-drop checks across the pipeline.